1 The Group 17 elements are oxidising agents.

(a) **(i)** Explain how the Group 17 elements act as oxidising agents.

.....

..... [1]

(ii) Write an equation to show the reaction in which Cl_2 oxidises aluminium metal.

..... [1]

2 Aluminium oxide, Al_2O_3, and phosphorus(V) oxide, P_4O_{10}, are both used as reagents and catalysts.

(a) The melting point of Al_2O_3 is 2072 °C. The melting point of P_4O_{10} is 340 °C.

Explain the difference in the melting points of these two compounds.

.....

.....

.....

..... [3]

(c) P_4O_{10} catalyses the reversible reaction of CO with H_2 to form CH_3OH.

equation 1 $\quad CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$

P_4O_{10} then acts as a dehydrating agent, causing CH_3OH to form CH_3OCH_3.

(i) Explain how the presence of a catalyst affects a chemical reaction.

.....

.....

..... [1]

(ii) Construct an equation for the dehydration reaction of CH_3OH to form CH_3OCH_3.

..... [1]

(iii) Write an equation to show the reaction of P_4O_{10} with an excess of water.

..... [1]

3 Propan-2-ol, $(CH_3)_2CHOH$, is sometimes added to fuel to help it burn.

Fig. 3.1 shows some reactions of propan-2-ol.

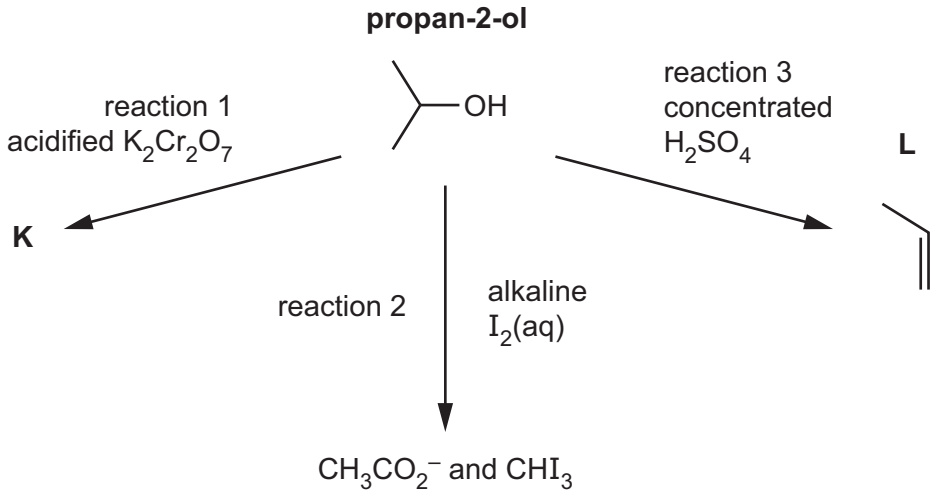


Fig. 3.1

(a) (i) Draw the structure of organic compound **K**.

[1]

(b) Propan-2-ol reacts with sodium to produce $(CH_3)_2CH–O^-$ anions.

These anions react with 2-bromopropane to form compound **N**, as shown in Fig. 3.3.

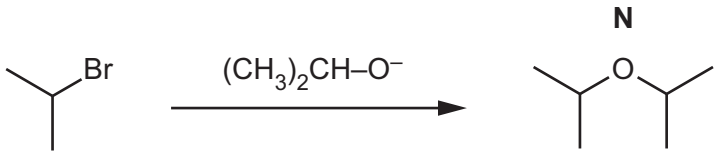


Fig. 3.3

(i) Write an equation for the reaction of propan-2-ol with sodium.

..... [1]

(iv) **N** is also added to petrol to make it burn more smoothly.

Construct an equation for the complete combustion of **N**, $C_6H_{14}O$.

$C_6H_{14}O$ ..... [1]

4 Compounds **P** and **Q** are structural isomers.



Fig. 4.1

(a) (i) Define structural isomerism.

.....

..... [2]

(iv) **Q** is oxidised by hot concentrated acidified $KMnO_4(aq)$, forming two different organic products.

Construct an equation for this reaction. Use [O] to represent an atom of oxygen from the oxidising agent.

$(CH_3)_2C{=}CHCH_3$ ..... [2]

(b) Fig. 4.3 shows a synthesis starting from **T**, a different isomer of **R** and **S**.

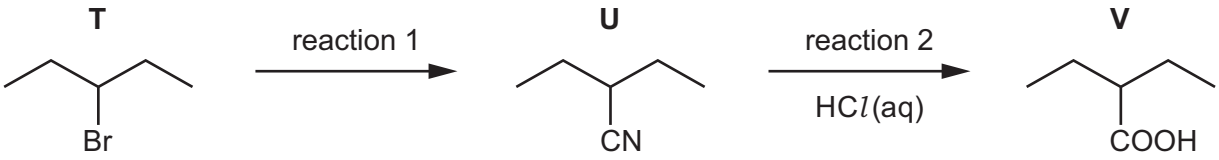


Fig. 4.3

(i) Identify the reagent and conditions for reaction 1.

..... [1]

(ii) Reaction 2 is a hydrolysis reaction.

Construct an equation for reaction 2.

$(C_2H_5)_2CHCN$ ..... [1]

1 Chromium, Cr, and its compounds are widely used in many chemical reactions.

(a) Cr exists as four stable isotopes.

(i) The most common isotope of Cr is chromium-52.

Determine the number of protons, neutrons and electrons in an atom of chromium-52.

number of protons ..... neutrons ..... electrons .....
[1]

(iii) The relative isotopic masses of the isotopes of Cr can be determined using mass spectrometry.

State what other information is needed to calculate the relative atomic mass, A_r, of Cr.

..... [1]

(iv) Atoms of ^{52}Cr, ^{53}Cr and ^{54}Cr make up more than 95% of naturally occurring chromium atoms. The A_r of naturally occurring Cr is 51.996.

Suggest what these statements imply about the relative isotopic mass of the fourth stable isotope of chromium.

..... [1]

(c) Acidified dichromate(VI) ions will convert methanal, CH_2O, to carbon dioxide. The movement of electrons to or from relevant species is shown in the following half-equations.

half-equation 1 $CH_2O + H_2O \rightarrow CO_2 + 4H^+ + 4e^-$

half-equation 2 $Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$

(i) Identify the species that is reduced in half-equation 2. Explain your answer.

..... [1]

(iii) Construct the ionic equation for the reaction of dichromate(VI) ions with methanal in acidic conditions.

..... [2]

(iii) CrO_4^{2-}(aq) ions are yellow and $Cr_2O_7^{2-}$(aq) ions are orange.

State what is observed when the following changes are made to an equilibrium mixture of acidified CrO_4^{2-}(aq) and $Cr_2O_7^{2-}$(aq) ions.

Explain your answers.

- The equilibrium mixture is warmed gently.

  observation .....

  explanation .....

  .....

  .....

- Dilute HC*l*(aq) is added to the equilibrium mixture.

  observation .....

  explanation .....

  .....

  .....

[4]

(e) Chromium(IV) fluoride, CrF_4, is a covalent molecule that shows similar chemical properties to SiCl_4.

Suggest the type of reaction that occurs when CrF_4 is placed in water.

Construct a relevant equation for this reaction.

type of reaction .....

equation .....

[2]

2 The Period 3 elements show trends in physical and chemical properties across the period.

(a) Fig. 2.1 shows the variation in atomic and ionic radii of the Period 3 elements Na to C*l*.

The ionic radius of Si is **not** shown.

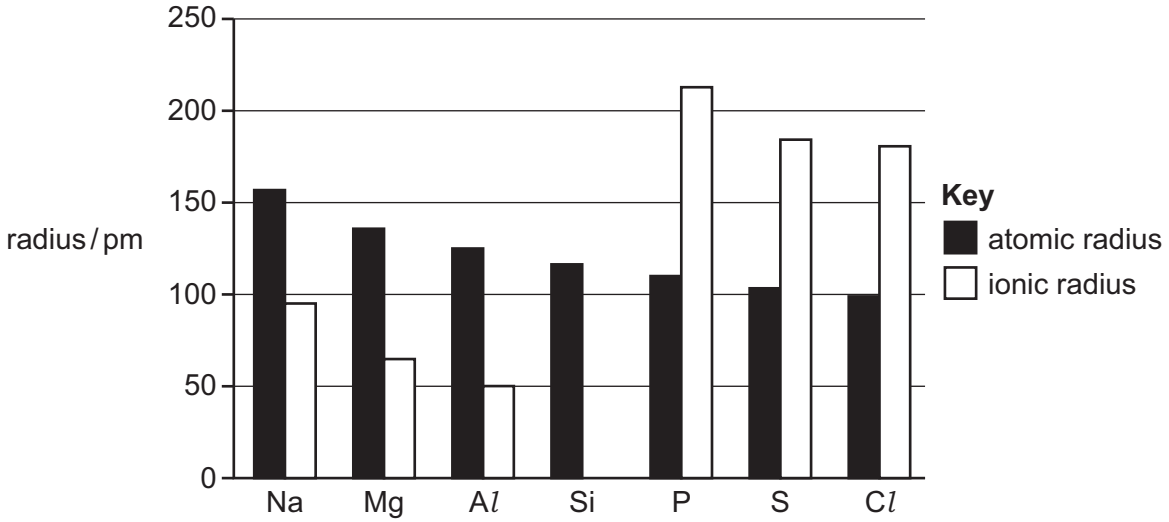


Fig. 2.1

(i) Explain the trend shown in the atomic radii of the Period 3 elements Na to C*l*.

.....

.....

.....

..... [2]

(c) **(i)** Write an equation for the reaction of Na_2O with dilute hydrochloric acid.

..... [1]

(ii) Construct an equation for the reaction of Al_2O_3 with a base to form $NaAlO_2$.

..... [1]

3 Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

(a) Define empirical formula.

.....

..... [1]

1 Manganese, Mn, and its compounds are widely used in many chemical reactions.

(a) Mn is usually found as a single isotope, manganese-55.

(i) Determine the number of protons, neutrons and electrons in an atom of manganese-55.

number of protons ..... neutrons ..... electrons .....
[1]

(ii) Define isotopes.

.....

..... [1]

(iii) A sample of manganese from the Moon is found to contain manganese-53 in addition to manganese-55.

State the two pieces of information needed to determine the relative atomic mass, A_r, of manganese in this sample.

1 .....

2 .....
[2]

(c) Manganese(IV) oxide reacts with methanal, CH_2O, in acidic conditions to produce carbon dioxide. The movement of electrons to or from relevant species is shown in the following half-equations.

half-equation 1 $CH_2O + H_2O \rightarrow CO_2 + 4H^+ + 4e^-$

half-equation 2 $MnO_2 + 4H^+ + 2e^- \rightarrow Mn^{2+} + 2H_2O$

(i) Identify the species that is reduced in half-equation 2. Explain your answer.

.....

..... [1]

(iii) Construct the ionic equation for the reaction of manganese(IV) oxide with methanal in acidic conditions.

..... [2]

2 The Period 3 elements show trends in physical and chemical properties across the period.

(a) (i) Explain why the elements Na to A*l* are good electrical conductors.

.....

..... [1]

(d) (i) Write an equation for the formation of A*l*C*l*$_3$ from its elements.

.....

..... [1]

(ii) Write an equation for the formation of H_3PO_4 from PC*l*$_5$.

..... [1]

(iii) S_2Cl_2(l) is yellow and SCl_2(l) is red.

$$S_2Cl_2(l) + Cl_2(g) \rightleftharpoons 2SCl_2(l) \qquad \Delta H = -41\,kJ\,mol^{-1}$$

State what is observed when the following changes are made to an equilibrium mixture of S_2Cl_2(l) and SCl_2(l).

Explain your answers.

- The equilibrium mixture is warmed gently.

  observation .....

  explanation .....

  .....

  .....

- The overall pressure of the equilibrium mixture is increased.

  observation .....

  explanation .....

  .....

  .....

[4]

(f) Aqueous MgC*l*$_2$ reacts with aqueous Na_2CO_3 to form a white precipitate. Upon heating, the white precipitate undergoes thermal decomposition.

(i) Construct an equation to show the reaction of aqueous MgC*l*$_2$ with aqueous Na_2CO_3. Use state symbols in your equation.

..... [2]

3 The alkanes are a homologous series of organic molecules. Alkanes are generally unreactive and are commonly used as fuels.

(a) Define homologous series.

.....

..... [2]

(iii) Write an equation for the complete combustion of hexane.

..... [1]

4 Fig. 4.1 shows how propane, C_3H_8, can be converted to propanoic acid, CH_3CH_2COOH.

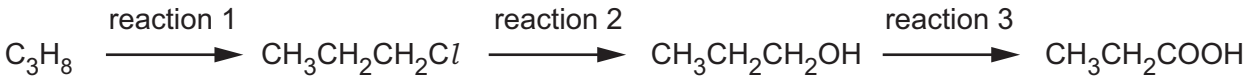


Fig. 4.1

(a) Reaction 1 in Fig. 4.1 takes place in the presence of sunlight.

$$C_3H_8 + Cl_2 \rightarrow CH_3CH_2CH_2Cl + HCl$$

The reaction takes place via initiation, propagation and termination steps.

(i) Name the mechanism shown by reaction 1.

..... [1]

(c) Reaction 3 takes place when $CH_3CH_2CH_2OH$ is heated under reflux with acidified potassium dichromate(VI) solution.

(i) State the colour change that takes place in the reaction mixture.

..... [1]

(ii) Construct an equation to represent reaction 3. Use [O] to represent an atom of oxygen from the oxidising agent.

..... [1]

1 Chromium, Cr, and its compounds are widely used in many chemical reactions.

(a) Cr exists as four stable isotopes.

(i) The most common isotope of Cr is chromium-52.

Determine the number of protons, neutrons and electrons in an atom of chromium-52.

number of protons ..... neutrons ..... electrons .....
[1]

(iii) The relative isotopic masses of the isotopes of Cr can be determined using mass spectrometry.

State what other information is needed to calculate the relative atomic mass, A_r, of Cr.

..... [1]

(iv) Atoms of ^{52}Cr, ^{53}Cr and ^{54}Cr make up more than 95% of naturally occurring chromium atoms. The A_r of naturally occurring Cr is 51.996.

Suggest what these statements imply about the relative isotopic mass of the fourth stable isotope of chromium.

..... [1]

(c) Acidified dichromate(VI) ions will convert methanal, CH_2O, to carbon dioxide. The movement of electrons to or from relevant species is shown in the following half-equations.

half-equation 1 $CH_2O + H_2O \rightarrow CO_2 + 4H^+ + 4e^-$

half-equation 2 $Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$

(i) Identify the species that is reduced in half-equation 2. Explain your answer.

..... [1]

(iii) Construct the ionic equation for the reaction of dichromate(VI) ions with methanal in acidic conditions.

..... [2]

(iii) CrO_4^{2-}(aq) ions are yellow and $Cr_2O_7^{2-}$(aq) ions are orange.

State what is observed when the following changes are made to an equilibrium mixture of acidified CrO_4^{2-}(aq) and $Cr_2O_7^{2-}$(aq) ions.

Explain your answers.

- The equilibrium mixture is warmed gently.

  observation .....

  explanation .....

  .....

  .....

- Dilute HC*l*(aq) is added to the equilibrium mixture.

  observation .....

  explanation .....

  .....

  .....

[4]

(e) Chromium(IV) fluoride, CrF_4, is a covalent molecule that shows similar chemical properties to SiCl_4.

Suggest the type of reaction that occurs when CrF_4 is placed in water.

Construct a relevant equation for this reaction.

type of reaction .....

equation .....

[2]

2 The Period 3 elements show trends in physical and chemical properties across the period.

(a) Fig. 2.1 shows the variation in atomic and ionic radii of the Period 3 elements Na to C*l*.

The ionic radius of Si is **not** shown.

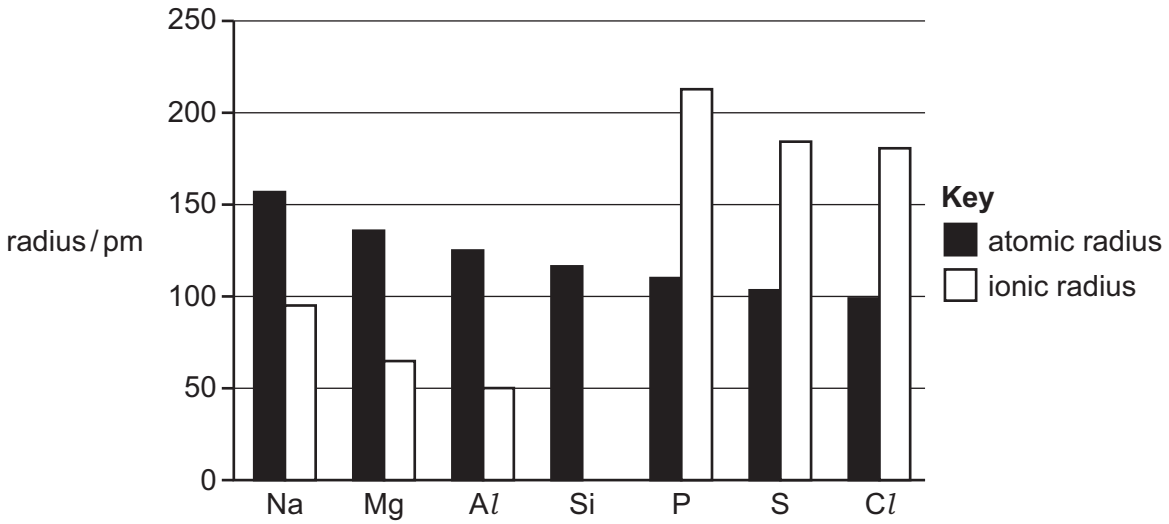


Fig. 2.1

(i) Explain the trend shown in the atomic radii of the Period 3 elements Na to C*l*.

.....

.....

.....

..... [2]

(c) **(i)** Write an equation for the reaction of Na_2O with dilute hydrochloric acid.

..... [1]

(ii) Construct an equation for the reaction of Al_2O_3 with a base to form $NaAlO_2$.

..... [1]

3 Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

(a) Define empirical formula.

.....

..... [1]

1 Elements in Period 3 of the Periodic Table show trends in their properties.

(a) Complete Table 1.1 by identifying the lattice structures of the crystalline solids of Mg, Si and P.

Table 1.1

element	Mg	Si	P
lattice structure in crystalline solid			

[3]

(c) The **third** ionisation energies of the Period 3 elements are shown in Fig. 1.2.

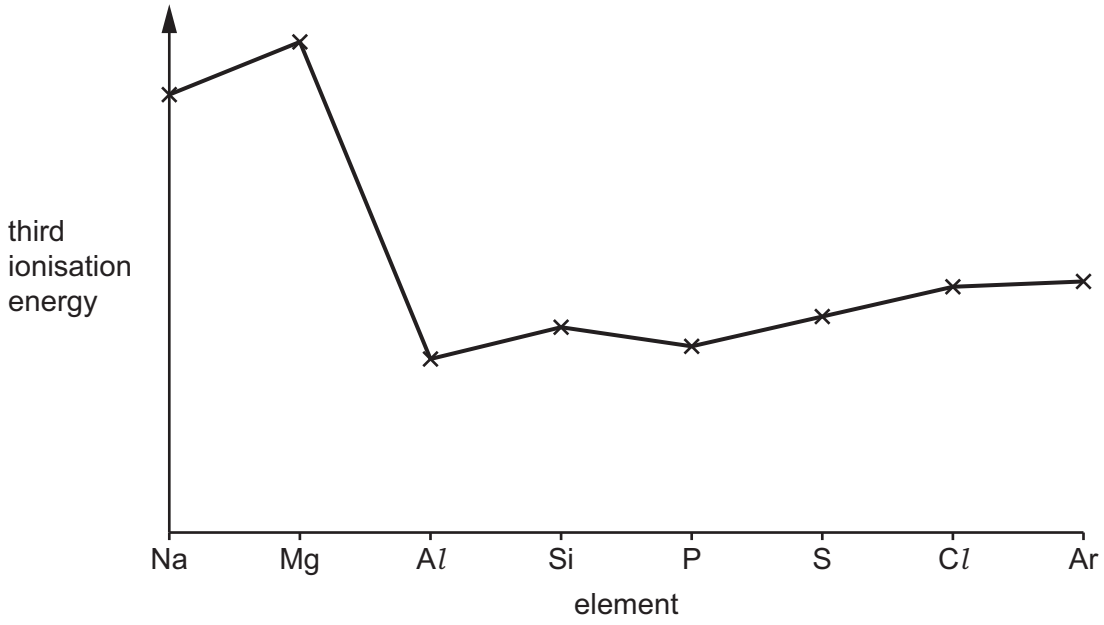


Fig. 1.2

(i) Write an equation, including state symbols, to represent the **third** ionisation energy of argon.

..... [2]

(d) Most Period 3 elements react with oxygen to form oxides.

(i) Write an equation for the reaction of phosphorus with oxygen.

..... [1]

(ii) Write an equation for the reaction of aluminium oxide with an excess of aqueous sodium hydroxide.

..... [1]

3 $H_2(g)$ and $I_2(g)$ react to form HI(g) in a reversible reaction, as shown in equation 1.

equation 1 $$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$$

In three separate experiments, a student combines different amounts of two or more of the gases from equation 1. In each experiment, the gases are left to reach equilibrium at a given temperature.

(a) In experiment 1, $H_2(g)$ and $I_2(g)$ are combined.

Water is then added to the equilibrium mixture to produce 1.00 dm^3 of solution **A**.

The amount of $I_2(aq)$ present in 1.00 dm^3 of solution **A** is found by titration with $Na_2S_2O_3(aq)$.

Exactly 32.90 cm^3 of 0.200 mol dm^{-3} $Na_2S_2O_3$ reacts with all of the I_2 in a 25.0 cm^3 sample of solution **A**.

$$I_2(aq) + 2S_2O_3^{2-}(aq) \rightarrow 2I^-(aq) + S_4O_6^{2-}(aq)$$

(i) Calculate the amount, in mol, of $S_2O_3^{2-}$ that reacts in the titration.

amount of $S_2O_3^{2-}$ = ... mol [1]

(ii) Use your answer to **(a)(i)** to calculate the amount, in mol, of I_2 present in 1.00 dm^3 of solution **A**.

(If you were unable to obtain an answer in **(a)(i)**, then use 9.42×10^{-3} mol. This is **not** the correct answer.)

amount of I_2 in 1.00 dm^3 of solution **A** = ... mol [2]

(b) In experiment 2, 0.100 mol of $H_2(g)$, 0.200 mol of $I_2(g)$ and 0.300 mol of HI(g) are combined.

At equilibrium 0.154 mol of I_2 is present.

Calculate the amounts, in mol, of H_2 and HI in the equilibrium mixture produced in experiment 2.

amount of H_2 in equilibrium mixture = ... mol

amount of HI in equilibrium mixture = ... mol [3]

4 The skeletal formula of **E** is shown in Fig. 4.1.

E

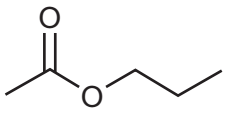


Fig. 4.1

(a) State the molecular formula of **E**.

..... [1]

(b) Name **E**.

..... [1]

(c) **E** is made when a carboxylic acid and an alcohol react together.

(i) Name the type of reaction that takes place during this esterification reaction.

..... [1]

(d) **F** is an isomer of **E**.
F reacts with HC*l*(aq) to produce **G** and **H**.
G is a secondary alcohol.
H is also produced when HC*l*(aq) is added to CH_3CN.

(i) Construct an equation for the acid hydrolysis of CH_3CN with HC*l*(aq).

..... [1]

6 **P**, **Q** and **R** are three different hydrocarbon molecules.

(a) (i) Define empirical formula.

.....

..... [1]

(ii) Hydrocarbon **P** contains 85.7% by mass of carbon.

Calculate the empirical formula of **P**.

Show your working.

empirical formula of **P** = ... [2]

2 The chemical properties of oxides are related to the chemical bonding present in these compounds.

(a) A Period 3 oxide produces a solution with a pH greater than 10 when it is added to water.
State the formula of the oxide.

..... [1]

(b) P_4O_{10} is added to an excess of aqueous NaOH.
Write an equation to describe the reaction.

..... [1]

(e) Al_2O_3 is a white amphoteric compound.

(i) State the formula of the aluminium-containing species produced when Al_2O_3 reacts with NaOH(aq).

..... [1]

(ii) State the formula of the aluminium-containing salt produced when Al_2O_3 reacts with H_2SO_4(aq).

..... [1]

4 The reaction between $Na_2S_2O_3$(aq) and HCl(aq) is monitored at constant temperature.

$$Na_2S_2O_3 + 2HCl \rightarrow 2NaCl + SO_2 + S + H_2O$$

Fig. 4.1 shows how the concentration of HCl(aq) varies with time.

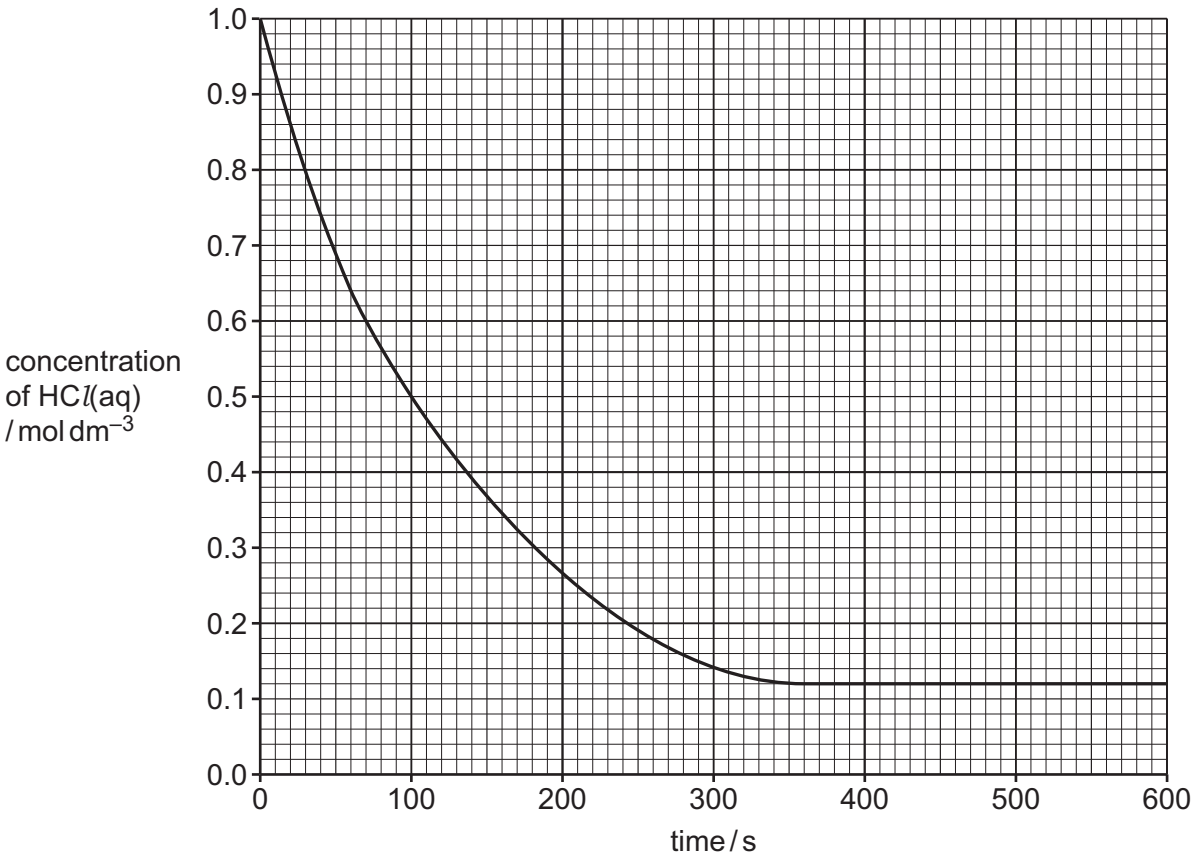


Fig. 4.1

(a) (i) Use Fig. 4.1 to find the average rate of change of concentration of HCl(aq) in this reaction between 0–100 seconds and between 400–500 seconds. Include units in your answers.

0–100 seconds ..... units .....

400–500 seconds ..... units .....

[2]

(ii) Use Fig. 4.1 to identify the limiting reagent. Explain your answer.

.....

..... [1]

(iv) The reaction between $Na_2S_2O_3(aq)$ and $HCl(aq)$ is repeated in a second experiment.

In this second experiment, 25.0 cm^3 of 0.050 mol dm^{-3} $Na_2S_2O_3(aq)$ reacts with 0.0020 mol of $HCl(aq)$. Calculate the number of sulfur atoms produced.

number of sulfur atoms produced = ... [2]

6 **(a)** $HOCl$ reacts with $CH_2{=}CH_2$ to produce $HOCH_2CH_2Cl$ in an electrophilic addition reaction.

(i) Define addition reaction.

.....

..... [1]

(c) $HOCH_2CH_2Cl$ reacts in a two-step synthesis to produce $HOCH_2CH_2COOH$, as shown in Fig. 6.2.

$$HOCH_2CH_2Cl \xrightarrow{\text{step 1}} HOCH_2CH_2CN \xrightarrow[HCl(aq)]{\text{step 2}} HOCH_2CH_2COOH$$

Fig. 6.2

(i) State the reagent and conditions for step 1.

..... [1]

(ii) Identify the type of reaction that occurs in step 2.

..... [1]

(iii) Complete the equation to show the reaction in step 2.

$HOCH_2CH_2CN$ + ... [1]

1 **(a)** Diamond and graphite are both crystalline solids made from carbon atoms. Graphite conducts electricity. Diamond does **not** conduct electricity.

(i) Name the type of lattice structure present in the crystalline solids diamond and graphite.

..... [1]

(iii) Explain why diamond does **not** conduct electricity.

.....

..... [1]

(b) Separate samples of phosphorus(V) chloride and silicon(IV) chloride are each added to an excess of cold water.

(i) Write an equation for each reaction.

.....

..... [2]

2 A sample of iron contains three different isotopes and has a relative atomic mass, A_r, of 55.8.

(a) (i) Define relative atomic mass.

.....

.....

..... [2]

(ii) Table 2.1 shows the abundances of two of the isotopes present in the sample of iron.

Table 2.1

isotope	relative isotopic mass	abundance / %
^{54}Fe	53.9	6.0
^{56}Fe	55.9	91.9

Use Table 2.1 to calculate the relative isotopic mass of the third isotope of iron in the sample. Show your working.

relative isotopic mass = ... [2]

3 **(a)** C_3H_6 reacts with HBr(g) in an addition reaction.

(i) Define addition reaction.

.....

..... [1]

(b) H_2(g) and nickel are added to alkene **X**. Fig. 3.2 shows how the concentration of **X** changes with time.

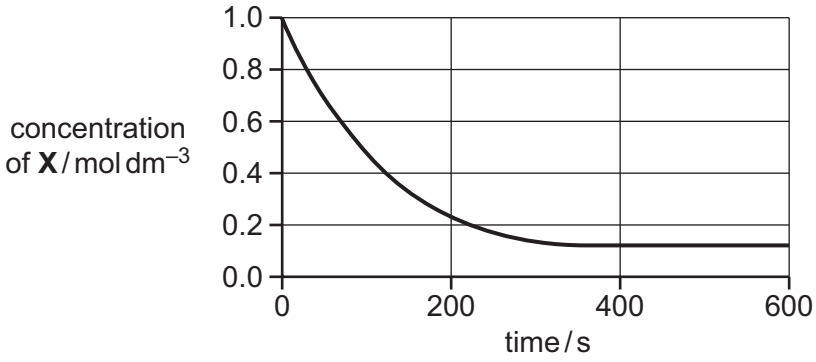


Fig. 3.2

(i) Use Fig. 3.2 to identify the limiting reagent in this reaction. Explain your answer.

.....

..... [1]

4 **(a)** Propanoic acid, CH_3CH_2COOH, reacts with reducing agent **Q** to produce propan-1-ol, $CH_3CH_2CH_2OH$.

(i) Suggest the formula of reducing agent **Q**.

..... [1]

(b) Propan-1-ol is converted to compound **T** in a three-step synthesis.

In step 1, $SOCl_2$ is added to propan-1-ol to produce compound **E**.

In step 2, **E** reacts with a suitable reagent to produce butanenitrile.

In step 3, butanenitrile is heated with NaOH(aq).

(i) Name the type of reaction that occurs in step 1.

..... [1]

(iii) Construct an equation to describe the reaction of butanenitrile with NaOH(aq) in step 3.

..... [2]

5 The structure of vitamin C is shown in Fig. 5.1.

vitamin C

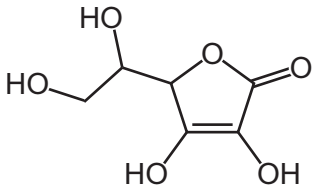


Fig. 5.1

(a) Deduce the empirical formula of vitamin C.

..... [1]

(b) The mass of vitamin C present in 150.0 g of lemon is found in an experiment.

stage 1 All the vitamin C in 150.0 g of lemon is extracted and dissolved in water to make 100.0 cm^3 of solution **L**.

stage 2 A 25.0 cm^3 sample of solution **L** is titrated with 5.00×10^{-3} mol dm^{-3} $I_2(aq)$.

vitamin C **M**

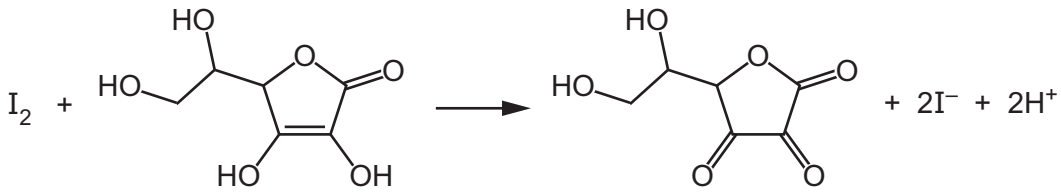


Fig. 5.2

Exactly 36.65 cm^3 of $I_2(aq)$ reacts with the 25.0 cm^3 sample of solution **L**.

[M_r: vitamin C, 176]

(i) Calculate the amount, in mol, of $I_2(aq)$ added in the titration.

amount of $I_2(aq)$ = ... mol [1]

(ii) Use your answer to **(b)(i)** to calculate the percentage by mass of vitamin C present in 150.0 g of lemon.

(If you were unable to calculate a value for the amount of I_2(aq) in **(b)(i)**, use the value 7.65×10^{-4} mol. This is **not** the correct value.)

percentage by mass = ...% [2]

1 **(a)** Solid sodium conducts electricity. Sodium oxide conducts electricity when molten but not when solid.

(i) Name the type of bonding present in sodium and in sodium oxide.

bonding in sodium .....

bonding in sodium oxide .....
[1]

(iii) Explain why sodium oxide conducts electricity when molten.

.....

..... [1]

(b) Separate samples of sodium and sodium oxide are each added to an excess of cold water.

(i) Write an equation for the reaction of sodium with cold water.

..... [1]

4 The structure of vitamin C is shown in Fig. 4.1.

vitamin C

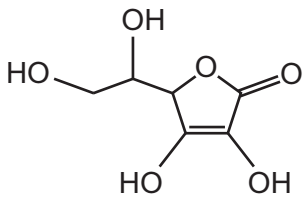


Fig. 4.1

(a) Deduce the empirical formula of vitamin C.

..... [1]

(b) The concentration of vitamin C is found by titration with $I_2(aq)$.

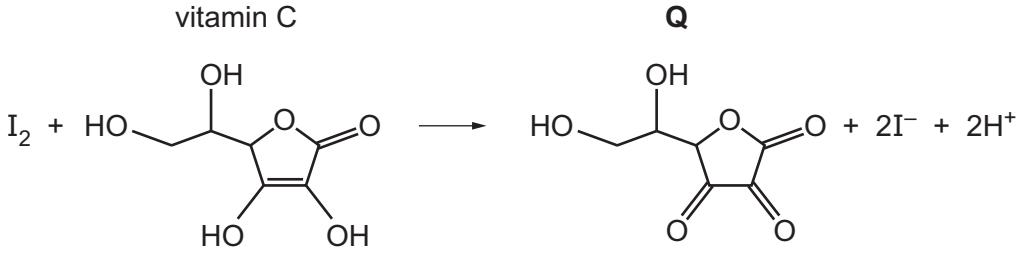


Fig. 4.2

A vitamin C tablet is dissolved in water to produce 200.0 cm^3 of vitamin C solution.
5.00 cm^3 of this vitamin C solution is added to a flask with approximately 150 cm^3 of water and an indicator.
Exactly 28.40 cm^3 of 5.00×10^{-4} mol dm^{-3} $I_2(aq)$ reacts with the sample of vitamin C solution in the flask.

[M_r: vitamin C, 176]

(i) Calculate the amount, in mol, of $I_2(aq)$ added to the flask in this titration.

amount of $I_2(aq)$ = ... mol [1]

(ii) Use your answer to **(b)(i)** to calculate the mass, in g, of vitamin C in the tablet. Show your working.

(If you were unable to calculate a value for the amount of I_2(aq) in **(b)(i)**, use the value 2.64×10^{-4} mol. This is **not** the correct value.)

mass of vitamin C in tablet = ...g [2]

[Total: 9]

Table 4.1

bond	functional groups containing the bond	characteristic infrared absorption range (in wavenumbers) / cm^{-1}
C−O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C−H	alkane	2850–2950
N−H	amine, amide	3300–3500
O−H	carboxyl hydroxy	2500–3000 3200–3650

5 Compound **A** contains the elements carbon, hydrogen and oxygen only.
When **A** is heated with $H_2SO_4(aq)$, compounds **B** and **C** are produced, as shown in Fig. 5.1.

$$\mathbf{A} + 2H_2O \xrightarrow[\text{heat}]{H_2SO_4(aq)} \underset{\mathbf{B}}{HOOCCH_2COOH} + \underset{\mathbf{C}}{2CH_3CH_2OH}$$

Fig. 5.1

(a) Draw the structure of **A**.

[2]

(c) When aqueous Na_2CO_3 is added to separate samples of **B** and **C**, effervescence is observed with **B** only.

(i) Complete the equation to describe the reaction between **B** and an excess of aqueous Na_2CO_3.

.....$HOOCCH_2COOH$ + ...Na_2CO_3 → .....
[1]

(e) **B** reacts with an excess of reducing agent **R** to produce **D**.

An excess of PCl_5 is added to **D**.

- A vigorous reaction occurs.
- Misty fumes are seen.
- Organic compound **E** is produced.

(i) Identify the formula of the reducing agent **R**.

..... [1]

(ii) Complete the equation to describe the reaction of **B** with an excess of **R**.
Use [H] to represent one atom of hydrogen from **R**.

.....$HOOCCH_2COOH$ + ...[H] → ..... [2]

1 Phosphorus and chlorine are elements in Period 3 of the Periodic Table.

(a) Chlorine forms three different compounds with phosphorus.

The most common compounds are PCl_3 and PCl_5.

(i) Complete Table 1.1.

Table 1.1

compound	oxidation number of P	oxidation number of Cl
PCl_3		
PCl_5		

[2]

(iii) The third compound of phosphorus and chlorine, **W**, has a relative molecular mass, M_r, between that of PCl_3 and PCl_5. The compound contains 69.6% by mass of chlorine.

Determine the molecular formula of **W**.

molecular formula of **W** = ... [2]

2 **(a)** The Group 2 elements show metallic bonding.

Define metallic bonding.

.....

..... [2]

(ii) Write an equation for the reaction of strontium, Sr, with an excess of cold water.

..... [1]

1 Cobalt, rhodium and iridium are metals in the same group of the Periodic Table.

(a) The shorthand electronic configuration of cobalt is [Ar]$3d^7 4s^2$.

(i) Identify what is meant by [Ar] by giving its full electronic configuration.

..... [1]

(c) Table 1.2 shows the relative abundances of isotopes in a sample of an alloy containing rhodium and iridium only.

Table 1.2

isotope	relative isotopic mass	relative abundance in alloy
$^{103}_{45}Rh$	102.91	50.00
$^{191}_{77}Ir$	190.96	15.18
$^{193}_{77}Ir$	192.96	34.82

(i) Define relative isotopic mass.

.....

.....

..... [2]

(ii) Use Table 1.2 to calculate the relative atomic mass, A_r, of iridium in the alloy.

Give your answer to **two** decimal places.

relative atomic mass of iridium = ... [2]

(d) Hydrated rhodium(III) chloride, $RhCl_3 \cdot xH_2O$, catalyses the conversion of ethene to but-2-ene.

Both stereoisomers of but-2-ene are formed in the reaction.

(i) Hydrated rhodium(III) chloride contains 20.5% by mass of water of crystallisation.

Deduce the integer value of x in $RhCl_3 \cdot xH_2O$.

Show your working.

x = ...
[2]

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2 Chlorine is one of the elements in Group 17 of the Periodic Table.

(a) (i) Describe the colours of the Group 17 elements, chlorine to iodine, at room temperature.

.....

..... [1]

(ii) State the essential condition for chlorine to react with C_2H_6 at room temperature.

..... [1]

(iii) Write numbers in the boxes to balance the equation showing the reaction of propanone with NaC*l*O.

CH_3COCH_3 + ☐ NaC*l*O → ☐ CHCl_3 + ☐ CH_3COONa + ☐ NaOH

[1]

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

(a) Table 3.1 shows some properties of two Group 14 elements, C and Sn, in their standard states. The table is incomplete.

Table 3.1

	C (graphite)	Sn
state and appearance in standard state	grey shiny solid	silvery solid
electrical conductivity		good
type of bonding		metallic
type of structure	giant	

(i) Complete Table 3.1. [3]

(b) Carbon is found in inorganic compounds such as carbonates.

(i) Write an equation for the reaction of magnesium carbonate with dilute HCl(aq).

..... [1]

(e) Silicon readily reacts with elements of high electronegativity.

(i) Write an equation for the formation of $SiCl_4$ from its constituent elements.

..... [1]

(f) Tin forms an amphoteric oxide, SnO_2.

Suggest the formula of the tin compound that forms when SnO_2 reacts with H_2SO_4 in an acid–base reaction.

..... [1]

1 Vanadium, niobium and tantalum are metals in the same group of the Periodic Table.

(a) The shorthand electronic configuration of vanadium in the ground state is $[Ar]3d^3 4s^2$.

(i) State what is meant by the term ground state.

..... [1]

(b) Pelopium was the suggested name for a new element discovered in a mineral.

Pelopium was later found to be a mixture of niobium, Nb, and tantalum, Ta.

Only one naturally occurring isotope exists for each of Nb and Ta.

(i) Complete Table 1.1.

Table 1.1

isotope	relative isotopic mass	number of protons	number of neutrons
$^{93}_{41}Nb$	92.91		
$^{181}_{73}Ta$	180.95		

[2]

(ii) Define relative isotopic mass.

.....

.....

..... [2]

(iii) A sample of pelopium contains 90.9% by mass $^{93}_{41}Nb$ and 9.1% by mass $^{181}_{73}Ta$.

Calculate the theoretical relative atomic mass of pelopium based on these data and Table 1.1.

Give your answer to two decimal places.

Show your working.

theoretical relative atomic mass of pelopium = ...
[2]

2 Oxygen is a Group 16 element.

(a) **(i)** Write equations for the following reactions.

- sodium and oxygen

.....

- sulfur and oxygen

.....

[2]

(ii) Write an equation for the reaction of CH_3CN with H_2O in acidic conditions.

CH_3CN + ...H_2O + ...H^+ .....

[1]

3 Nitrogen and phosphorus are elements in Group 15 of the Periodic Table.

(a) Nitrogen is found in inorganic compounds such as nitrogen oxides (NO_x), nitrates and nitric acid.

(i) Identify **one** natural and **one** man-made occurrence of nitrogen oxides in the atmosphere.

natural .....

man-made .....
[2]

(ii) Write an equation to describe the role of NO_2 in the direct formation of acid rain.

..... [1]

(iv) Nitric acid reacts with basic oxides to form nitrates.

Write an equation for the reaction of nitric acid with calcium oxide.

..... [1]

(v) Describe what is seen when solid calcium nitrate is heated strongly.

..... [1]

1 Cobalt, rhodium and iridium are metals in the same group of the Periodic Table.

(a) The shorthand electronic configuration of cobalt is [Ar]$3d^7 4s^2$.

(i) Identify what is meant by [Ar] by giving its full electronic configuration.

..... [1]

(c) Table 1.2 shows the relative abundances of isotopes in a sample of an alloy containing rhodium and iridium only.

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$^{103}_{45}Rh$	102.91	50.00
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(i) Define relative isotopic mass.

.....

.....

..... [2]

(d) Hydrated rhodium(III) chloride, $RhCl_3 \cdot xH_2O$, catalyses the conversion of ethene to but-2-ene.

Both stereoisomers of but-2-ene are formed in the reaction.

(i) Hydrated rhodium(III) chloride contains 20.5% by mass of water of crystallisation.

Deduce the integer value of x in $RhCl_3 \cdot xH_2O$.

Show your working.

x = ...
[2]

BLANK PAGE

2 Chlorine is one of the elements in Group 17 of the Periodic Table.

(a) (i) Describe the colours of the Group 17 elements, chlorine to iodine, at room temperature.

.....

..... [1]

(iii) State what is observed when chlorine reacts with hydrogen.

..... [1]

(ii) State the essential condition for chlorine to react with C_2H_6 at room temperature.

..... [1]

(iii) Write **two** equations to show the propagation steps in this reaction.

1 .....

2 .....
[2]

(iii) Write numbers in the boxes to balance the equation showing the reaction of propanone with NaC*l*O.

CH_3COCH_3 + ☐ NaC*l*O → ☐ CHCl_3 + ☐ CH_3COONa + ☐ NaOH
[1]

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

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type of bonding		metallic
type of structure	giant	

(i) Complete Table 3.1. [3]

(e) Silicon readily reacts with elements of high electronegativity.

(i) Write an equation for the formation of $SiCl_4$ from its constituent elements.

..... [1]

(f) Tin forms an amphoteric oxide, SnO_2.

Suggest the formula of the tin compound that forms when SnO_2 reacts with H_2SO_4 in an acid–base reaction.

..... [1]

1 **(a)** **(i)** Explain the lack of reactivity of nitrogen gas, $N_2(g)$.

.....

.....

..... [2]

(c) Separate samples of aluminium oxide, Al_2O_3, and phosphorus(V) oxide, P_4O_{10}, react with an excess of NaOH(aq) at room temperature.

(i) Give the state of Al_2O_3 and P_4O_{10} at room temperature.

Al_2O_3 .....

P_4O_{10} .....
[1]

(ii) Write an equation for the reaction of each oxide with an excess of NaOH(aq) at room temperature.

Al_2O_3 + ...

P_4O_{10} + ...
[2]

2 $N_2(g)$ reacts with $H_2(g)$ in the Haber process, as shown in reaction 1.

reaction 1 $\quad N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) \quad \Delta H = -x\ kJ\,mol^{-1}$

Table 2.1 shows the different conditions used to produce three equilibrium mixtures, **A**, **B** and **C**.

Table 2.1

	A	**B**	**C**
initial molar ratio of N_2 : H_2 added	1 : 3	1 : 3	1 : 3
temperature / °C	500	500	1000
pressure / atm	1000	1000	1000
iron present in mixture	no	yes	no
percentage yield of $NH_3(g)$ at equilibrium	58	x	y

(a) Describe and explain the change, if any, to the percentage yield of $NH_3(g)$ produced in **B** compared to **A**.

.....

.....

..... [1]

(d) When oxides of nitrogen escape into the atmosphere they may be involved in:

- formation of acid rain from sulfur dioxide
- formation of photochemical smog.

(i) Identify the role of NO and NO_2 in the formation of H_2SO_4 from SO_2 in the atmosphere to produce acid rain.

Use relevant equations to support your answer.

.....

.....

.....

.....

.....

..... [3]

3 **(a)** Write an equation to show the reaction for the standard enthalpy change of formation of H_2O. Include state symbols.

..... [2]

(c) Boron carbide is a hard crystalline solid that has a melting point greater than 2000 °C.

(i) Suggest the structure and bonding in boron carbide.

..... [1]

(ii) 100 g of pure boron carbide contains 78.26 g of boron.

Calculate the empirical formula of boron carbide.

Show your working.

empirical formula of boron carbide ..... [2]

4 **(a)** $NH_3(g)$ reacts with $HCl(g)$ to produce $NH_4Cl(s)$, as shown.

$$NH_3(g) + HCl(g) \rightarrow NH_4Cl(s)$$

Draw a diagram to show the ionic, covalent and coordinate bonding present in a formula unit of NH_4Cl.

[2]

(b) An exothermic reaction occurs when $NH_4^+(aq)$ is added to $OH^-(aq)$.

(i) Identify the type of reaction.

..... [1]

(ii) Construct an ionic equation for the reaction of NH_4^+ and OH^-.

..... [1]

1 **(a)** Complete Table 1.1 using relevant information from the Periodic Table.

Table 1.1

	nucleon number	proton number	number of electrons
Mg^{2+}	24		
Al^{3+}	27		

[2]

(iii) Write an equation to describe the reaction between P_4O_{10} and an excess of water.

..... [1]

(f) Aluminium hydroxide is amphoteric.

(i) Explain what is meant by amphoteric.

.....

..... [1]

(ii) Write an equation to describe the reaction that occurs when aluminium hydroxide, $Al(OH)_3$, reacts with NaOH(aq).

..... [1]

2 Separate samples of Na_2CO_3 and $NaHCO_3$ react with HC*l*(aq) to produce the same products, as shown in Table 2.1.

Table 2.1

reaction	equation	ΔH / kJ mol^{-1}
1	$Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2$	ΔH_1
2	$NaHCO_3 + HCl \rightarrow NaCl + H_2O + CO_2$	$\Delta H_2 = +27.2$

(a) Complete the reaction pathway diagram in Fig. 2.1 for reaction 2.

Label the diagram to show the enthalpy change, ΔH_2, and the activation energy, E_A.

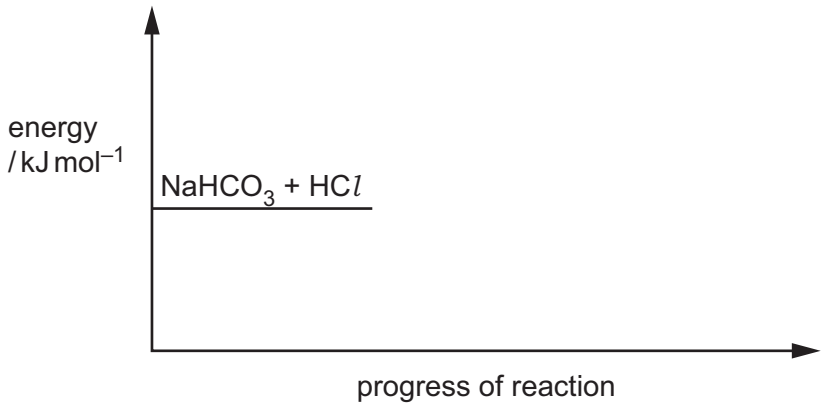


Fig. 2.1

[2]

(b) The value for ΔH_1 is determined by experiment using the following method.

- 50.0 cm^3 of 2.00 mol dm^{-3} HC*l*(aq) is added to a polystyrene cup.
- The initial temperature of the acid is recorded as 19.6 °C.
- 0.0400 mol of Na_2CO_3 is added and the mixture is stirred.
- All the solid Na_2CO_3 disappears and a colourless solution is produced.

The maximum temperature recorded during the reaction is 26.2 °C.

(i) Describe **one** other observation that shows the reaction is complete.

..... [1]

(ii) Calculate the value of ΔH_1 in $kJ\,mol^{-1}$.

Assume the specific heat capacity of the reaction mixture is the same as for water and no heat is lost to the surroundings.

Show your working.

ΔH_1 = ... $kJ\,mol^{-1}$ [3]

(c) **Z** is a salt that contains a Period 4 element from Group 2. When **Z** is heated brown gas forms.

Identify the formula of **Z** and use it to write an equation for the reaction.

..... [2]

5 Compound **W** has molecular formula $C_4H_{10}O$. It contains only **one** functional group.

(a) Table 5.1 shows the two peaks with the greatest *m*/*e* values in the mass spectrum of **W**.

Table 5.1

m/*e*	relative abundance
74	50
75	x

(i) Calculate the relative abundance, x, of the peak at *m*/*e* = 75 using the information from Table 5.1.

x = ... [1]

(c) **Y** is a structural isomer of **W**.

Both **W** and **Y** produce colourless bubbles when sodium is added to them.

Y does **not** react when heated with acidified $K_2Cr_2O_7$.

Y does **not** react when warmed with alkaline I_2(aq).

(i) Name the functional group present in **Y**.

..... [1]

(ii) Complete the equation to describe the reaction of **W** or **Y** with sodium.

.....$C_4H_{10}O$ + ...Na → ..... [1]

1 **(a)** The elements of Group 17 are called halogens.

Complete Table 1.1.

Table 1.1

halogen	colour at 293 K
chlorine	
bromine	
iodine	

[1]

(c) Iodine is made by reacting bromine with sodium iodide.

(i) Construct an ionic equation for the reaction of bromine with sodium iodide.

..... [1]

(d) Concentrated sulfuric acid is added to separate samples containing equal amounts of NaCl, NaBr and NaI. All three samples initially react to produce the hydrogen halide.

(i) Write an equation to describe the acid–base reaction that occurs when concentrated sulfuric acid reacts with NaBr.

..... [1]

2 **(a)** Sulfur chloride, SCl_2, is a liquid at room temperature. When SCl_2 is added to water, misty fumes are seen and a solution is made that turns universal indicator red.

(i) Identify the type of reaction that occurs when SCl_2 is added to water.

..... [1]

(c) Solid magnesium nitride, Mg_3N_2, is a crystalline solid.

(i) Deduce the oxidation numbers of magnesium and nitrogen in magnesium nitride to complete Table 2.1.

Table 2.1

	oxidation number in Mg_3N_2
magnesium	
nitrogen	

[1]

(ii) Magnesium nitride reacts with an excess of water to produce ammonia and magnesium hydroxide only. Construct an equation to describe this reaction.

..... [1]

(d) Boron nitride is a white solid that melts above 2900 °C.

Fig. 2.2 shows part of the lattice structure of a crystal of boron nitride.

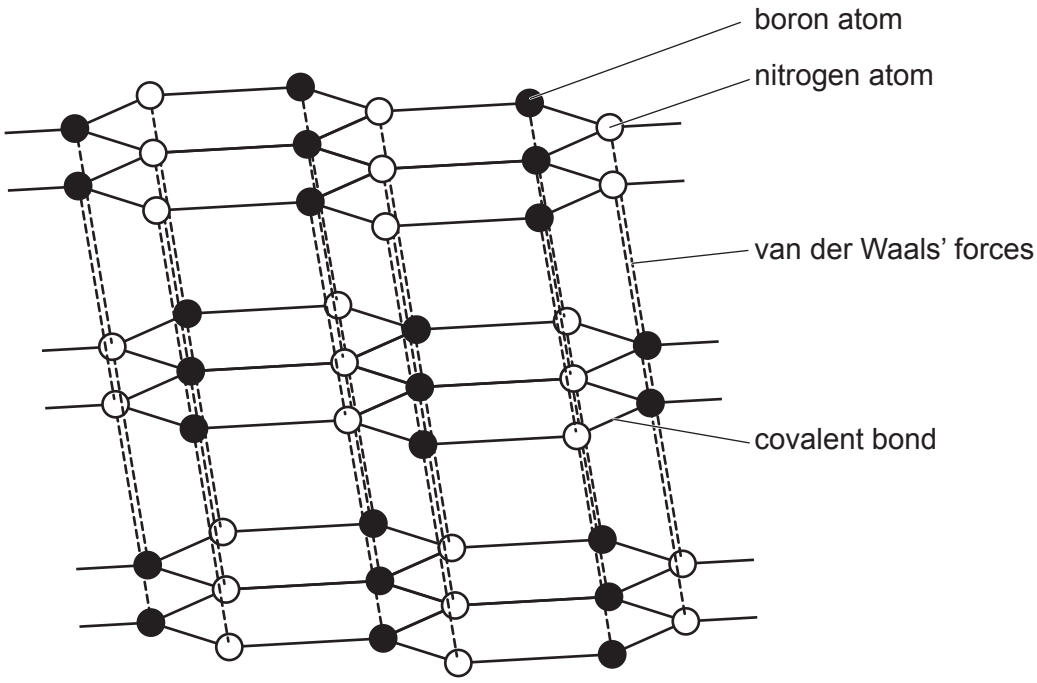


Fig. 2.2

(i) Use Fig. 2.2 to deduce the empirical formula of boron nitride.

..... [1]

3 **(a)** Define Le Chatelier's principle.

.....

.....

..... [2]

(c) In another experiment, equimolar amounts of $Fe^{3+}(aq)$ and $SCN^{-}(aq)$ are mixed together and allowed to reach equilibrium. The total volume of the mixture is 25.0 cm^3.

reaction 1 $Fe^{3+}(aq) + SCN^{-}(aq) \rightleftharpoons FeSCN^{2+}(aq)$

At equilibrium the mixture contains:

- $[SCN^{-}] = 1.30 \times 10^{-3}\,mol\,dm^{-3}$
- $[FeSCN^{2+}] = 0.300 \times 10^{-3}\,mol\,dm^{-3}$.

(i) Calculate the initial amount, in mol, of $Fe^{3+}(aq)$ added to $SCN^{-}(aq)$ to produce this mixture.

initial amount of $Fe^{3+}(aq)$ = ... mol [2]

4 **(a)** Define enthalpy change of formation.

.....

.....

..... [2]

(b) Iron is made when iron(III) oxide is heated with carbon monoxide, as shown by reaction 2.

reaction 2 $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$

Table 4.1 shows enthalpy change of formation data measured at 298 K and 101 kPa.

Table 4.1

substance	equation	value for $\Delta H_f^{\ominus}$ / kJ mol^{-1}
Fe_2O_3		–824.2
CO		–110.5
CO_2	$C(s) + O_2(g) \rightarrow CO_2(g)$	–393.5

(i) Complete Table 4.1 by adding equations with relevant state symbols to represent:

- standard enthalpy change of formation for Fe_2O_3
- standard enthalpy change of formation for CO.

[2]

6 **(a)** **V** shows stereoisomerism.

V

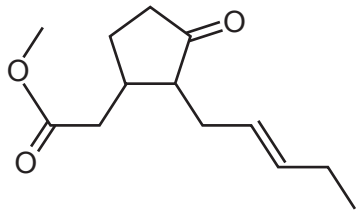


Fig. 6.1

(i) Explain what is meant by stereoisomerism.

.....

.....

..... [1]

(c) Both functional groups in one molecule of **Y** react with an inorganic reagent to form one molecule of **Q** and one molecule of methanol, CH_3OH, as shown in Fig. 6.3.

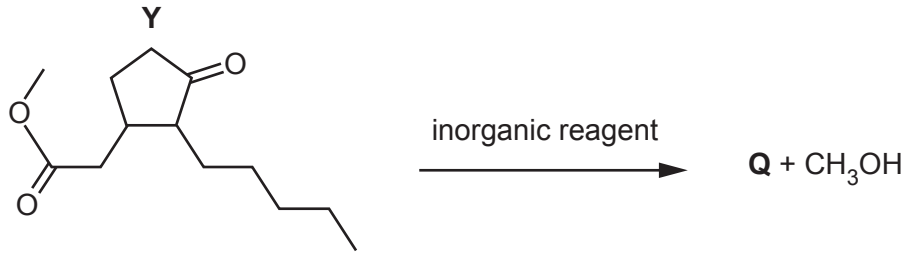


Fig. 6.3

(i) Part of the mass spectrum for **Q** is shown in Fig. 6.4. Only peaks with *m*/*e* greater than 198 are shown.

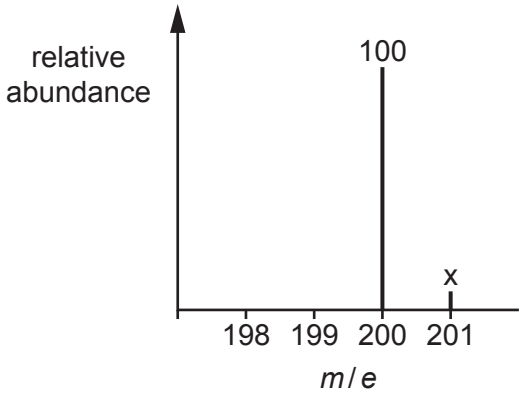


Fig. 6.4

Calculate the relative abundance, x, of the peak at *m*/*e* = 201.

Show your working.

x = ... [2]

(ii) **Q** contains **only** hydroxyl functional groups.

Complete Table 6.1 to show the observations that occur when 2,4-dinitrophenylhydrazine (2,4-DNPH reagent) is added to separate samples of **Y** and **Q**.

Table 6.1

	observation on addition of 2,4-DNPH reagent
Y	
Q	

[1]

(iii) Under certain conditions, 0.0020 mol of **Q** reacts with an excess of sodium to produce a total of 44.8 cm^3 of gas at s.t.p.

Calculate the number of hydroxyl groups present in a molecule of **Q**.

Show your working.

number of hydroxyl groups = ... [2]

1 Bismuth is an element in Group 15 of the Periodic Table.

(a) Bismuth has metallic bonding.

Draw a labelled diagram to show the metallic bonding in bismuth.

[1]

(b) Bismuth reduces water to form bismuth oxide, Bi_2O_3. A colourless gas that ignites with a squeaky pop also forms.

(i) Construct an equation for the reduction of water by bismuth.

..... [1]

(e) $NaBiO_3$ can be used to determine the concentration of $Mn^{2+}(aq)$. The ionic equation for the reaction is shown in equation 2.

equation 2 $\quad 2Mn^{2+} + 5BiO_3^- + 14H^+ \rightarrow 2MnO_4^- + 5Bi^{3+} + 7H_2O$

A student uses the following procedure in an experiment.

- Add 100.0 cm^3 of a saturated solution of $Mn^{2+}(aq)$ to a volumetric flask.
- Add distilled water to the flask to make a 1.00 dm^3 diluted solution.
- Titrate a 25.00 cm^3 sample of the diluted solution with 0.100 mol dm^{-3} $NaBiO_3(aq)$.

The 25.00 cm^3 sample of the diluted solution of $Mn^{2+}(aq)$ reacts completely with exactly 21.50 cm^3 of 0.100 mol dm^{-3} $NaBiO_3(aq)$.

Calculate the concentration, in mol dm^{-3}, of $Mn^{2+}(aq)$ in the saturated solution.

Show your working.

concentration of $Mn^{2+}(aq)$ in the saturated solution = ... mol dm^{-3} [3]

2 Chlorine, Cl_2, reacts with many elements and compounds to form chlorides.

Table 2.1 shows information about some chlorides of Period 3 elements.

Table 2.1

	Na	Mg	Si
formula of chloride			
structure of chloride	giant		
bonding of chloride			covalent
pH of solution formed on addition of chloride to water		6.2	

(a) Complete Table 2.1. [3]

(b) When Cl_2 reacts with **cold** NaOH(aq), Cl_2 is both oxidised and reduced. The products are NaCl, water and **G**.

(i) State the type of redox reaction in which the same species is both oxidised and reduced.

..... [1]

(iii) Write an equation for the reaction between Cl_2 and **hot** NaOH(aq).

..... [1]

3 Nitrogen, N_2, is generally an unreactive molecule but it does react under certain conditions.

(a) Give **two** reasons to explain the lack of reactivity of nitrogen.

1 .....

2 .....
[2]

(b) N_2 can react with oxygen in an internal combustion engine to form a mixture of NO and NO_2. Fig. 3.1 shows a reaction scheme involving N_2.

reaction 1	N_2	$\xrightarrow{O_2}$	NO and NO_2
reaction 2	NO_2	$\xrightarrow{H_2O}$	products
reaction 3	NO_2	$\xrightarrow{\text{unburned hydrocarbons}}$	peroxyacetyl nitrate (PAN)

Fig. 3.1

(i) Write an equation to show the formation of a mixture of NO and NO_2 in reaction 1.

..... [1]

4 Compound **S** is used in food flavourings. A possible synthesis of **S** is shown in Fig. 4.1.

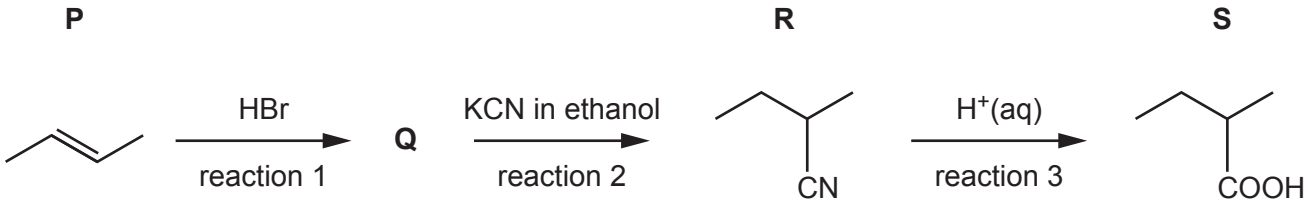


Fig. 4.1

(a) **P**, **Q**, **R** and **S** show stereoisomerism.

Complete Table 4.1 by identifying with a tick (✓) the type of stereoisomerism that each molecule shows.

The type of stereoisomerism shown by **Q** is given.

Table 4.1

	P	**Q**	**R**	**S**
geometrical isomerism				
optical isomerism		✓		

[2]

(iii) Complete the equation for reaction 3. **R** is represented as C_4H_9CN.

C_4H_9CN + ... [1]

1 The elements phosphorus, sulfur and chlorine are in Period 3 of the Periodic Table.

Table 1.1 shows some properties of the elements P to C*l*.

The first ionisation energy of S is **not** shown.

Table 1.1

property	P	S	C*l*
number of electrons in 3p subshell			
total number of unpaired electrons			
first ionisation energy /kJ mol^{-1}	1060		1260
formula of most common anion	P^{3-}	S^{2-}	Cl^{-}

(a) **(i)** Complete Table 1.1 to show the number of electrons in the 3p subshell and the total number of unpaired electrons in an atom of P, S and C*l*. [2]

(d) $POCl_3$ shows similar chemical properties to PCl_5.

$POCl_3$ has a melting point of 1°C and a boiling point of 106 °C.

$POCl_3$ reacts vigorously with water, forming misty fumes and an acidic solution.

(i) Explain how the information in **(d)** suggests the structure and bonding of $POCl_3$ is simple covalent.

.....

.....

..... [2]

(ii) Construct an equation for the reaction of $POCl_3$ with water.

$POCl_3$ + ... → .....
[1]

2 Barium hydroxide, $Ba(OH)_2$, is a strong base used in inorganic and organic reactions.

Fig. 2.1 shows a reaction scheme involving $Ba(OH)_2$.

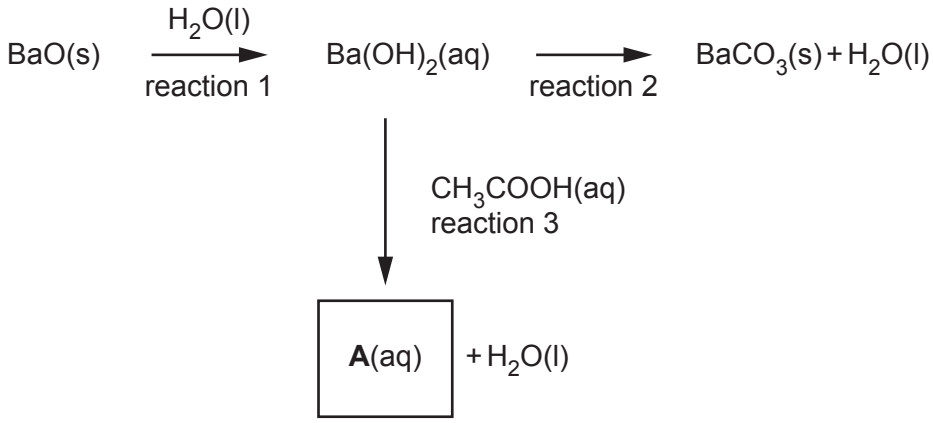


Fig. 2.1

(a) (i) State the variation in solubilities of group 2 hydroxides.

..... [1]

(iv) Identify **A**.

..... [1]

(v) $Ba(OH)_2$ is made by the reaction of Ba with water.

Write an equation for this reaction.

..... [1]

(b) The mineral barytocalcite contains both $BaCO_3$ and $CaCO_3$. Both compounds decompose on heating.

(i) State which compound decomposes first when barytocalcite is heated.

Explain your answer.

.....

..... [1]

(ii) Construct an equation for the complete thermal decomposition of barytocalcite.

The formula of barytocalcite is $BaCa(CO_3)_2$.

$BaCa(CO_3)_2$ ..... [1]

3 Potassium chlorate, $KClO_3$, is widely used as an oxidising agent and to make $O_2(g)$.

(a) Define oxidising agent.

.....

..... [1]

(d) Molten $KClO_3$ reacts with glucose, $C_6H_{12}O_6$.

$$4KClO_3 + C_6H_{12}O_6 \rightarrow 6CO_2 + 6H_2O + 4KCl$$

$KClO_3$ melts at 630 K. At this temperature, both CO_2 and H_2O are gases.

(i) Use the ideal gas equation to calculate the volume, in m^3, of one mole of gas at 630 K and 1.00×10^5 Pa.

Show your working. Give your answer to 3 significant figures.

volume of 1 mol of gas = ... m^3
[1]

(ii) 5.00 g of $C_6H_{12}O_6$ reacts completely with molten $KClO_3$.

Use your answer to **(d)(i)** to calculate the total volume of gas released at 630 K and 1.00×10^5 Pa in this reaction.

(If you were unable to answer **(d)(i)**, use 0.0463 m^3 in this question. This is **not** the correct answer to **(d)(i)**.)

total volume of gas released = ... m^3
[2]

2 NO and NO_2 react at 25 °C to give N_2O_3 as shown in the equation.

$$NO(g) + NO_2(g) \rightleftharpoons N_2O_3(g) \qquad \Delta H = -7.2\,kJ\,mol^{-1}$$

The reaction is reversible and reaches equilibrium in a closed system.

(a) Fig. 2.1 shows how the rate of the forward reaction changes with time.

Initially, the rate of the reverse reaction is zero.

Complete Fig. 2.1 to sketch how the rate of the **reverse** reaction changes with time.

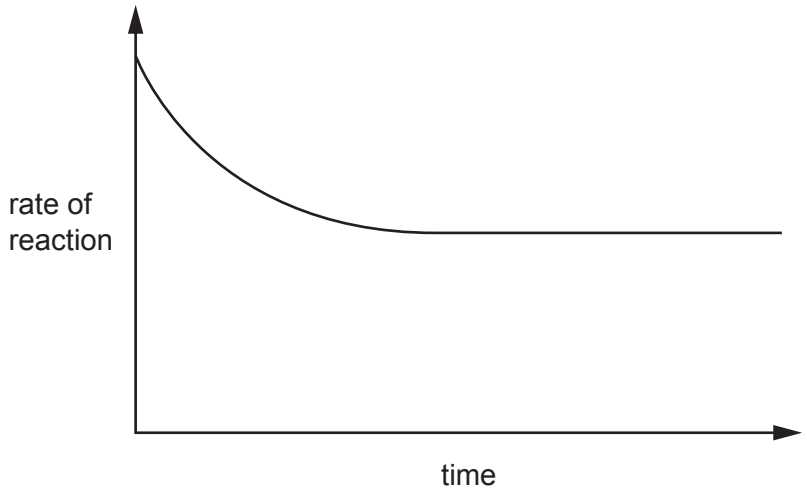


Fig. 2.1

[1]

(e) NO_2 is a brown gas that can be used to form nitric acid.

(i) NO_2 is a free radical.

Define free radical.

..... [1]

(ii) NO_2 has a catalytic role in the oxidation of atmospheric sulfur dioxide.

Write equations to show the catalytic role of NO_2 in this oxidation.

.....

..... [2]

(f) A student titrates nitric acid with a base to form a solution containing aqueous magnesium nitrate.

(i) Identify a base that the student could use.

..... [1]

(ii) The student evaporates the water to obtain magnesium nitrate solid. When this solid is heated it decomposes.

Write an equation for the decomposition of magnesium nitrate.

..... [1]

3 Phosphoric(V) acid, H_3PO_4, is used in both inorganic and organic reactions.

(a) H_3PO_4 is made in a two-step process from phosphorus.

step 1 Phosphorus reacts with an excess of oxygen to form a white solid.

step 2 The white solid then reacts with water to form H_3PO_4.

(i) Write an equation for each step.

step 1 .....

step 2 .....

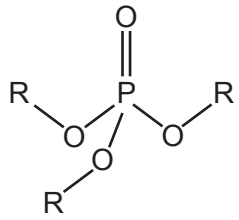
[2]

Table 3.2

bond	functional groups containing the bond	characteristic infrared absorption range (in wavenumbers) / cm^{-1}
C–O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C–H	alkane	2850–2950

(d) H_3PO_4 also reacts with alcohols to form organophosphates.

Organophosphates are compounds similar to esters. They have the general structure shown in Fig. 3.2.



R = alkyl group

Fig. 3.2

(i) Complete the equation to suggest the products of the reaction of H_3PO_4 with methanol, CH_3OH.

$H_3PO_4 + 3CH_3OH \rightarrow$ ..... [1]

4 Lactic acid, $CH_3CH(OH)COOH$, and pyruvic acid, $CH_3COCOOH$, both contain two functional groups.

lactic acid **pyruvic acid**

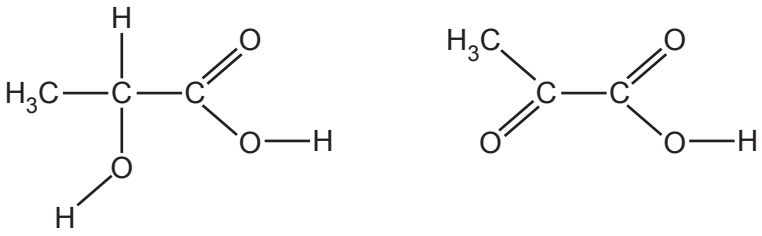


Fig. 4.1

(a) (i) Explain why lactic acid exists as optical isomers.

.....

.....

..... [1]

(b) Two possible syntheses of pyruvic acid are shown in Fig. 4.3 and Fig. 4.4.

Each synthesis has a total of three steps.

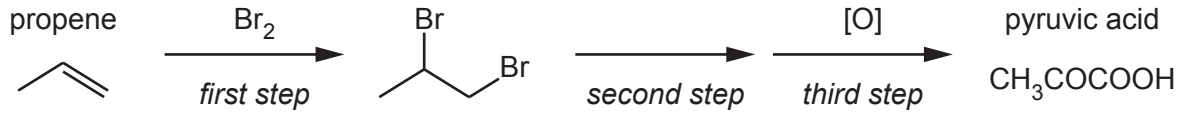


Fig. 4.3

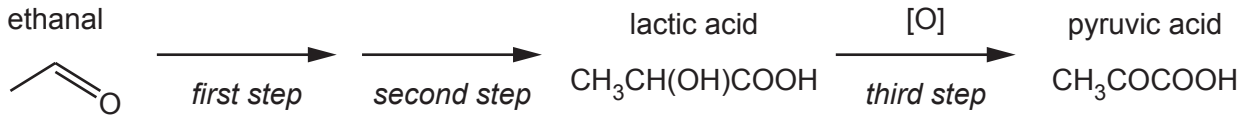


Fig. 4.4

(i) Complete the diagram in Fig. 4.5 to show the mechanism for the reaction of propene with Br_2.

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.



Fig. 4.5

[3]

(ii) Write an equation for the oxidation of lactic acid to pyruvic acid, the third step of Fig. 4.4.

Use [O] to represent one atom of oxygen from an oxidising agent.

$CH_3CH(OH)COOH$ + ... [1]

1 The elements phosphorus, sulfur and chlorine are in Period 3 of the Periodic Table.

Table 1.1 shows some properties of the elements P to C*l*.

The first ionisation energy of S is **not** shown.

Table 1.1

property	P	S	C*l*
number of electrons in 3p subshell			
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$POCl_3$ reacts vigorously with water, forming misty fumes and an acidic solution.

(i) Explain how the information in **(d)** suggests the structure and bonding of $POCl_3$ is simple covalent.

.....

.....

..... [2]

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$POCl_3$ + ... → .....
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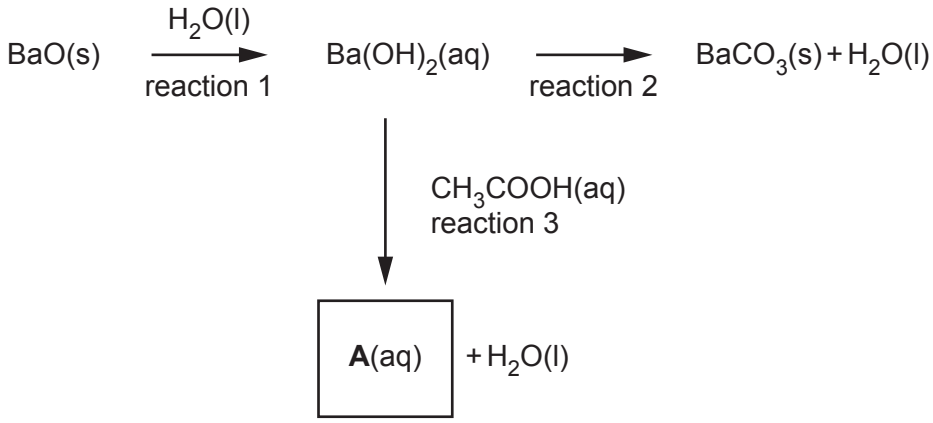


Fig. 2.1

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..... [1]

(iv) Identify **A**.

..... [1]

(v) $Ba(OH)_2$ is made by the reaction of Ba with water.

Write an equation for this reaction.

..... [1]

(b) The mineral barytocalcite contains both $BaCO_3$ and $CaCO_3$. Both compounds decompose on heating.

(i) State which compound decomposes first when barytocalcite is heated.

Explain your answer.

.....

..... [1]

(ii) Construct an equation for the complete thermal decomposition of barytocalcite.

The formula of barytocalcite is $BaCa(CO_3)_2$.

$BaCa(CO_3)_2$ ..... [1]

3 Potassium chlorate, $KClO_3$, is widely used as an oxidising agent and to make $O_2(g)$.

(a) Define oxidising agent.

.....

..... [1]

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$$4KClO_3 + C_6H_{12}O_6 \rightarrow 6CO_2 + 6H_2O + 4KCl$$

$KClO_3$ melts at 630 K. At this temperature, both CO_2 and H_2O are gases.

(i) Use the ideal gas equation to calculate the volume, in m^3, of one mole of gas at 630 K and 1.00×10^5 Pa.

Show your working. Give your answer to 3 significant figures.

volume of 1 mol of gas = ... m^3
[1]

(ii) 5.00 g of $C_6H_{12}O_6$ reacts completely with molten $KClO_3$.

Use your answer to **(d)(i)** to calculate the total volume of gas released at 630 K and 1.00×10^5 Pa in this reaction.

(If you were unable to answer **(d)(i)**, use 0.0463 m^3 in this question. This is **not** the correct answer to **(d)(i)**.)

total volume of gas released = ... m^3
[2]

1 Copper is used in electrical equipment. It has a melting point of 1085 °C.

(a) **(i)** Identify the lattice structure of copper.

..... [1]

(b) The relative isotopic masses and natural abundances of the two isotopes in a sample of copper are shown in Table 1.1.

Table 1.1

isotope	relative isotopic mass	% abundance
^{63}Cu	62.930	69.15
^{65}Cu	64.928	30.85

(i) Define the unified atomic mass unit.

.....

..... [1]

(ii) Define relative atomic mass, A_r, in terms of the unified atomic mass unit.

.....

..... [1]

(d) When KI(aq) is added to $CuSO_4$(aq) the blue-coloured solution turns brown and a white precipitate of CuI(s) is seen.

The reaction between copper ions and iodide forms only two products.

(i) Complete the equation for this reaction.

.....Cu^{2+} + ...I^- → .....CuI + ...

[1]

2 **(a)** The reaction of pure aluminium is only observed if the aluminium oxide layer is removed first. When pure aluminium is added to cold water, bubbles of gas are seen.

(i) State **one** property of aluminium oxide that explains why an aluminium object does **not** react with cold water until the aluminium oxide layer is removed.

..... [1]

(ii) Write an equation, with state symbols, for the reaction of aluminium oxide with an excess of NaOH(aq).

..... [2]

(b) Aluminium nitrate is a white soluble salt. On heating aluminium nitrate, thermal decomposition occurs and a brown gas is seen.

State the formula of the salt of another element in Period 3 which also decomposes on heating to produce a brown gas.

..... [1]

(c) Aluminium chloride and phosphorus chloride are both white solids.

(i) State the maximum oxidation number of aluminium and of phosphorus in these solid chloride salts.

maximum oxidation number of aluminium .....

maximum oxidation number of phosphorus .....

[1]

(iii) Write an equation for the reaction of solid phosphorus chloride and excess water.

..... [1]

3 A neutralisation reaction occurs when NaOH(aq) is added to H_2SO_4(aq).

equation 1 $2NaOH(aq) + H_2SO_4(aq) \rightarrow Na_2SO_4(aq) + 2H_2O(l)$

(a) Define enthalpy change of neutralisation, ΔH_{neut}.

.....

.....

..... [2]

(b) In an experiment, 50.0 cm^3 of 2.00 mol dm^{-3} NaOH(aq) is added to 60.0 cm^3 of 1.00 mol dm^{-3} H_2SO_4(aq) in a polystyrene cup and stirred. Both solutions have a temperature of 21.4 °C before mixing. The maximum temperature of the mixture is measured.

(i) Use equation 1 to calculate the amount, in mol, of H_2SO_4(aq) that is neutralised in the experiment.

amount of H_2SO_4 neutralised = ... mol [1]

(c) The enthalpy change of neutralisation of CH_3COOH(aq) and NaOH(aq) is –55.2 kJ mol^{-1}.

(i) Complete the equation for the reaction.

.....CH_3COOH + ...NaOH $\rightarrow$ ..... [1]

4 **(a)** Hydrogen chloride gas is made in the laboratory by adding concentrated sulfuric acid to potassium chloride.

(i) Construct an equation for this reaction.

..... [1]

5 **Y** is formed from **X** in a single-step reaction, as shown in Fig. 5.1.



Fig. 5.1

(a) Deduce the empirical formula of **Y**.

..... [1]

(b) The formation of **Y** from **X** requires the addition of a suitable reducing agent.

(i) Construct an equation using molecular formulae and [H] for the reaction in Fig. 5.1. Use [H] to represent one atom of hydrogen from the reducing agent.

..... [1]

6 Compound **W**, CH_2=CHCN, is used to make an addition polymer which is present in carbon fibres.

(a) Draw **one** repeat unit of the addition polymer of **W**.

[1]

(e) Propanenitrile is heated with hydrogen gas and a platinum catalyst. The only product is propylamine.

Construct an equation for this reaction.

..... [1]

2 Chlorine is a reactive element. It forms many compounds.

(a) **(i)** Complete Table 2.1 to show the maximum oxidation number of the elements Na to P in their chlorides.

Table 2.1

element	Na	Mg	A*l*	Si	P
maximum oxidation number					

[1]

(b) An excess of cold water is added to the chloride of silicon.

(i) Write an equation for the reaction between an excess of cold water and the chloride of silicon.

..... [1]

(c) An excess of cold water is added to the chloride of phosphorus.

(i) Write an equation for the reaction between an excess of cold water and the chloride of phosphorus.

..... [1]

(d) **(i)** Write an equation for the reaction of chlorine with water.

..... [1]

(ii) Write an equation for the reaction of chlorine with hot NaOH(aq).

..... [1]

(ii) Sodium chlorate(I), NaC*l*O, reacts with hydrogen peroxide to produce sodium chloride, water and oxygen gas.

Construct an equation for this reaction.

..... [1]

(iii) A sample of bleach **W** contains an unknown concentration of sodium chlorate(I).

10.0 cm^3 of **W** is diluted with distilled water to make a total volume of 100 cm^3 of bleach solution. 25.0 cm^3 of this diluted bleach solution is added to an excess of hydrogen peroxide and the volume of gas produced measured under room conditions.
The experiment is repeated and on average 25.0 cm^3 of diluted bleach solution produces 42.0 cm^3 of gas.

Calculate the concentration, in g dm^{-3}, of sodium chlorate(I) in **W**.

concentration of NaC*l*O in **W** = ... g dm^{-3} [3]

3 Fig. 3.1 describes a sequence of reactions that can be used to produce a food additive, compound **Y**, from CH_3CH_2Cl.

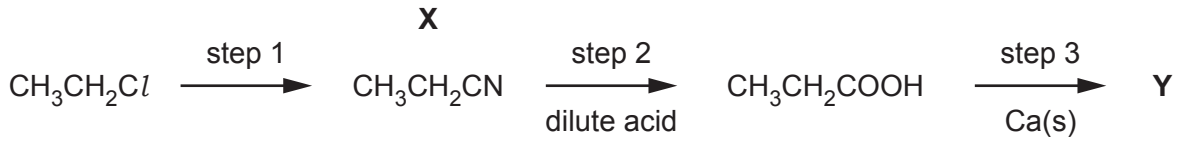


Fig. 3.1

(a) (i) State the reagent and conditions for step 1 in Fig. 3.1.

..... [1]

(iv) In step 3, **Y** and a gas are produced.
Construct an equation for step 3.

..... [2]

1 Tellurium is an element in Group 16. The most common isotope of tellurium is ^{130}Te. Its electronic configuration is [Kr] $4d^{10}\ 5s^2\ 5p^4$.

(a) Complete Table 1.1.

Table 1.1

	nucleon number	number of neutrons	number of electrons
^{130}Te			

[3]

(e) Te reacts with F_2 at 150 °C to form TeF_x. Molecules of TeF_x are octahedral with bond angles of 90°.

Explain why TeF_x is octahedral with bond angles of 90°.

.....

.....

..... [2]

(f) TeF_x reacts with water to form tellurium hydroxide and HF. The oxidation number of tellurium does **not** change during this reaction.

(i) Construct an equation for the reaction of TeF_x with water.

..... [1]

2 A neutralisation reaction occurs when NaOH(aq) is added to H_2SO_4(aq).

equation 1 $2NaOH(aq) + H_2SO_4(aq) \rightarrow Na_2SO_4(aq) + 2H_2O(l)$

(a) Define enthalpy change of neutralisation, ΔH_{neut}.

.....

.....

..... [2]

(b) An experiment is carried out to calculate ΔH_{neut} for the reaction between NaOH(aq) and H_2SO_4(aq).

100 cm^3 of 1.00 mol dm^{-3} NaOH(aq) is added to 75 cm^3 of 1.00 mol dm^{-3} H_2SO_4(aq) in a polystyrene cup and stirred. Results from the experiment are shown in Table 2.1.

Table 2.1

initial temperature of NaOH(aq) / °C	20.0
initial temperature of H_2SO_4(aq) / °C	20.0
maximum temperature of mixture / °C	27.8

(i) Use equation 1 to calculate the amount, in mol, of H_2SO_4(aq) that is neutralised in the experiment.

amount of H_2SO_4(aq) neutralised = ... mol [1]

(c) **(i)** Complete the equation for the reaction that occurs when a solution of $Ba(OH)_2$ is added to aqueous sulfuric acid. Include state symbols.

..... H_2SO_4(aq) + ...$Ba(OH)_2$(aq) → ……
[2]

3 Chlorine is a very reactive element.

(a) Chlorine reacts with silicon to form silicon(IV) chloride.
Describe the appearance of silicon(IV) chloride at room temperature and pressure. State its structure and bonding.

appearance .....

structure and bonding .....
[2]

(e) $CHCl_3$ and HF are used to form $CHClF_2$ in a substitution reaction.

Construct an equation for this reaction.

..... [1]

4 **V** is a colourless liquid.

V

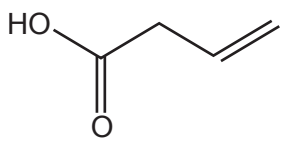


Fig. 4.1

(a) **V** reacts with an excess of $LiAlH_4$ to form **W**.

(i) Draw the structure of **W** in the box.

W

[1]

(b) **V** reacts to form **Z** in a single reaction, as shown in Fig. 4.2.

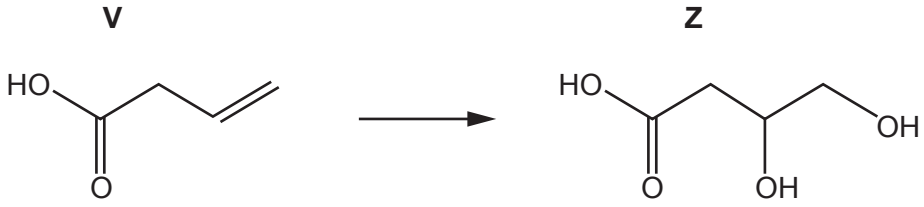


Fig. 4.2

(i) Suggest the reagent and conditions needed to form **Z** from **V**.

..... [1]

(ii) Deduce the empirical formula of **Z**.

..... [1]

5 **(a)** Molecule **M** is present in petrol, a fuel used in cars. **M** is a saturated, non-cyclic hydrocarbon. **M** contains eight carbon atoms.

(i) Construct an equation for the complete combustion of **M**.

..... [2]

(b) When petrol is burned in an internal combustion engine, oxides of nitrogen are released into the atmosphere. Oxides of nitrogen are responsible for the formation of acid rain.

(i) Suggest the conditions required for the production of oxides of nitrogen during combustion of **M** in an internal combustion engine. Use an appropriate equation in your answer.

.....

.....

.....

..... [2]

1 The Pauling electronegativity values of elements can be used to predict the chemical properties of compounds.

Use the information in Table 1.1 to answer the following questions.

Table 1.1

element	H	Li	C	O	S
Pauling electronegativity value	2.1	1.0	2.5	3.5	2.6
first ionisation energy / kJ mol^{-1}	1310	519	1090	1310	1000
second ionisation energy / kJ mol^{-1}	—	7300	2350	3390	2260

(a) (i) Define electronegativity.

.....

..... [1]

(d) CO_2 and SO_2 are acidic gases.

(i) Write an equation for the reaction of SO_2 with H_2O.

..... [1]

(ii) Write an equation for the reaction of SO_2 with NaOH.

..... [1]

(iii) Construct an equation for the reaction of CO_2 with $Mg(OH)_2$.

..... [1]

2 The Group 2 elements Mg to Ba are all silvery-white reactive metals.

(a) (i) Draw a labelled diagram to show the bonding and structure of the Group 2 metals at room temperature.

[2]

(b) Write an equation for the reaction of magnesium with cold water.

..... [1]

(ii) Compound **X**, an anhydrous Group 2 bromide, is dissolved in water and titrated against aqueous silver nitrate.

A solution containing 0.250 g of **X** requires 33.65 cm^3 of 0.0500 mol dm^{-3} $AgNO_3(aq)$ for complete reaction.

Identify **X**.

Show your working.

X = ... [3]

3 Alkenes undergo an addition reaction with a 1:1 mixture of CO and H_2 to form aldehydes.

Fig. 3.1 shows the reaction of propene with a 1:1 mixture of CO and H_2.

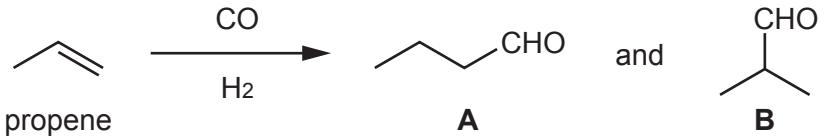


Fig. 3.1

(a) (i) Define addition reaction.

.....

..... [1]

(ii) Aldehydes **A** and **B** are structural isomers.

State the type of structural isomerism shown by **A** and **B**.

..... [1]

(iii) Name **A**.

..... [1]

(iv) The complete reaction of propene with a 1:1 mixture of CO and H_2 produces **A** and **B** only. The product mixture contains 96% **A** and 4% **B**.

Calculate the mass of **A** produced in this reaction when 5.00×10^3 kg of propene is used.

mass of **A** = ... kg [1]

(iii) **B**, C_4H_8O, is oxidised by acidified potassium manganate(VII).

Complete the equation for this reaction. Use [O] to represent one atom of oxygen from the oxidising agent.

C_4H_8O + ... [1]